

C PROGRAMMING : THREADING



The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named 'thread.c'. The code in the file is as follows:

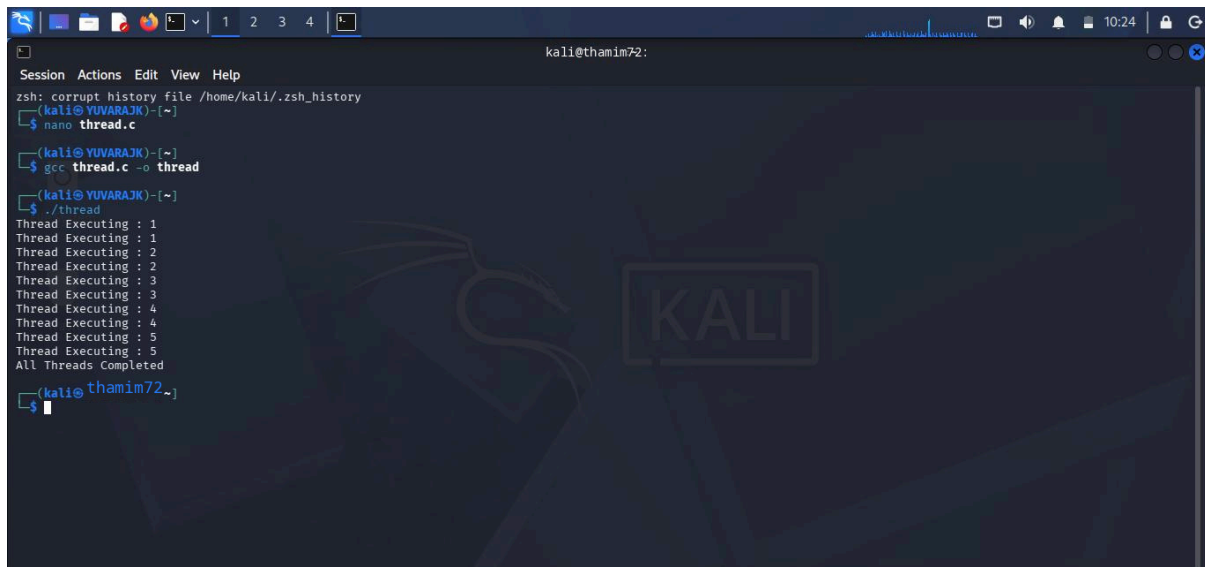
```
GNU nano 9.0 thread.c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

void *thread_function(void *arg)
{
    int i;
    for(i = 1; i <= 5; i++)
    {
        printf("Thread Executing : %d\n", i);
        sleep(1);
    }
    pthread_exit(NULL);
}

int main()
{
    pthread_t t1, t2;
    pthread_create(&t1, NULL, thread_function, NULL);
    pthread_create(&t2, NULL, thread_function, NULL);
    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    printf("All Threads Completed\n");
    return 0;
}
```

The terminal window title is 'kali@YUVARAJK: ~'. The nano editor's status bar at the bottom shows 'Read 23 lines'.

OUTPUT :

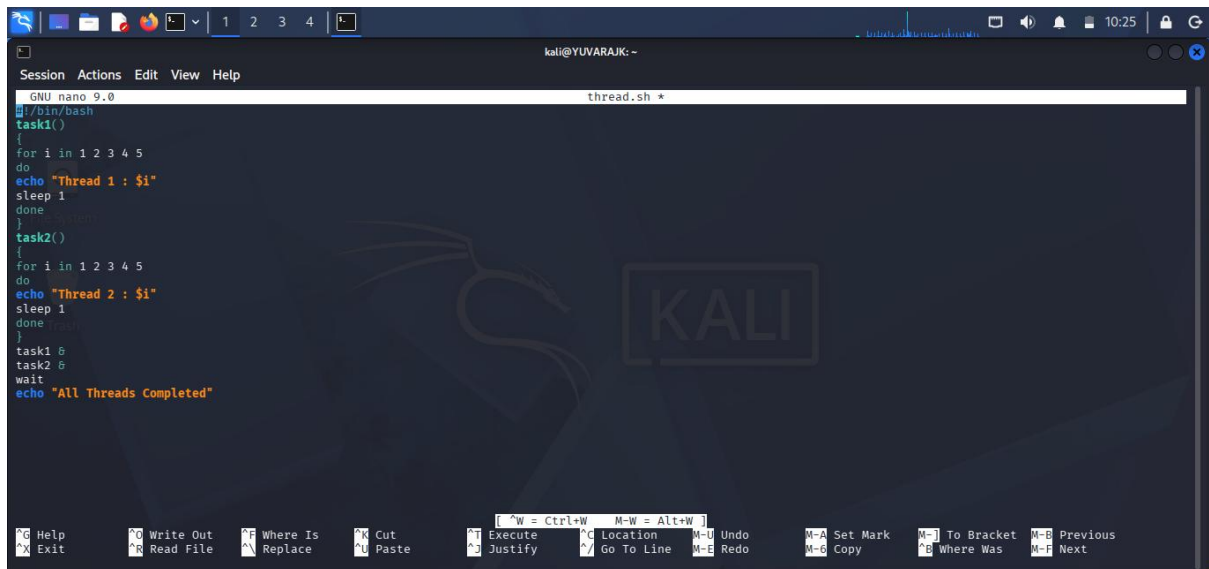


The screenshot shows a terminal window with the following commands and output:

```
kali@thamim72: ~
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)~$ nano thread.c
(kali@YUVARAJK)~$ gcc thread.c -o thread
(kali@YUVARAJK)~$ ./thread
Thread Executing : 1
Thread Executing : 1
Thread Executing : 2
Thread Executing : 2
Thread Executing : 3
Thread Executing : 3
Thread Executing : 4
Thread Executing : 4
Thread Executing : 5
Thread Executing : 5
All Threads Completed
(kali@thamim72)~$
```

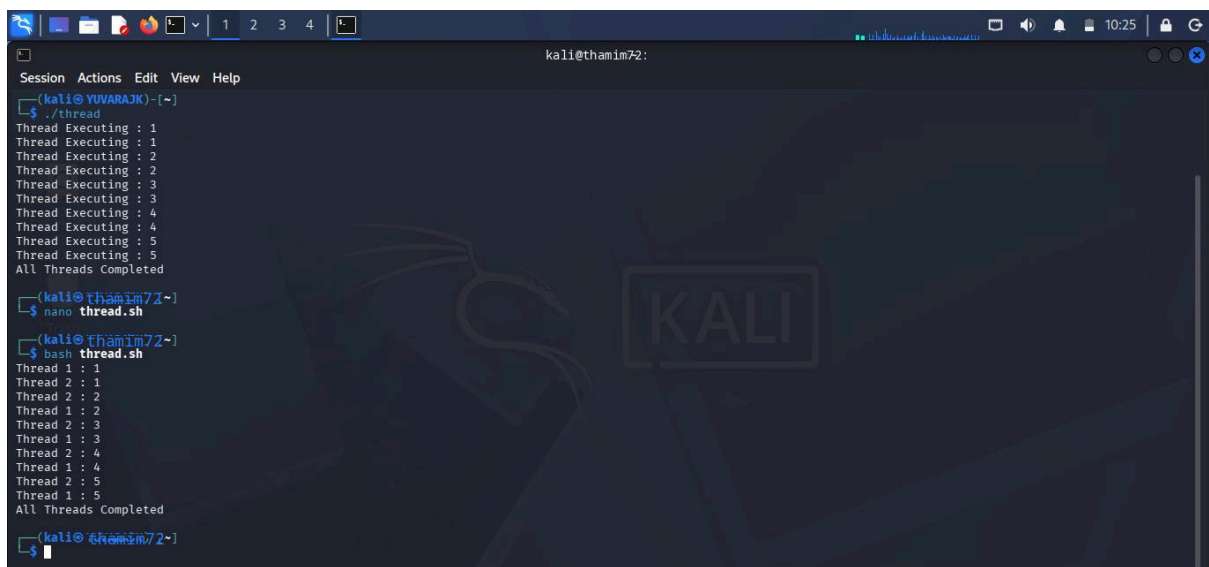
The terminal window title is 'kali@thamim72: ~'. The output shows five threads executing in parallel, each printing their ID and sleeping for 1 second. The program ends with 'All Threads Completed'.

SHELL PROGRAMMING : THREADING



```
GNU nano 9.0 thread.sh *
#!/bin/bash
task1()
{
for i in 1 2 3 4 5
do
echo "Thread 1 : $i"
sleep 1
done
}
task2()
{
for i in 1 2 3 4 5
do
echo "Thread 2 : $i"
sleep 1
done
}
task1 &
task2 &
wait
echo "All Threads Completed"
```

OUTPUT :



```
(kali@YUVARAJK)-[~]
$ ./thread
Thread Executing : 1
Thread Executing : 1
Thread Executing : 2
Thread Executing : 2
Thread Executing : 3
Thread Executing : 3
Thread Executing : 4
Thread Executing : 4
Thread Executing : 5
Thread Executing : 5
All Threads Completed

(kali@thamim72~)
$ nano thread.sh

(kali@thamim72~)
$ bash thread.sh
Thread 1 : 1
Thread 2 : 1
Thread 2 : 2
Thread 1 : 2
Thread 2 : 3
Thread 1 : 3
Thread 2 : 4
Thread 1 : 4
Thread 2 : 5
Thread 1 : 5
All Threads Completed

(kali@thamim72~)
$
```